

- **C₄ plants** minimize the cost of photorespiration by incorporating CO₂ into four-carbon compounds in mesophyll cells. These compounds are exported to **bundle-sheath cells**, where they release carbon dioxide for use in the Calvin cycle.
- **CAM plants** open their stomata at night, incorporating CO₂ into organic acids, which are stored in mesophyll cells. During the day, the stomata close, and the CO₂ is released from the organic acids for use in the Calvin cycle.
- Organic compounds produced by photosynthesis provide the energy and building material for Earth's ecosystems.

? Why are C₄ and CAM photosynthesis more energetically expensive than C₃ photosynthesis? What climate conditions would favor C₄ and CAM plants?

CONCEPT 10.6

Photosynthesis is essential for life on Earth: a review (pp. 206–209)

- Organic compounds produced by photosynthesis provide the energy and building material for Earth's ecosystems.

? How do plants use the products of photosynthesis?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- The light reactions supply the Calvin cycle with
 - light energy.
 - CO₂ and ATP.
 - H₂O and NADPH.
 - ATP and NADPH.
- Which of the following sequences correctly represents the flow of electrons during photosynthesis?
 - NADPH → O₂ → CO₂
 - H₂O → NADPH → Calvin cycle
 - H₂O → photosystem II → photosystem I
 - NADPH → electron transport chain → O₂
- How is photosynthesis similar in C₄ plants and CAM plants?
 - In both cases, only photosystem I is used.
 - Both types of plants make sugar without the Calvin cycle.
 - In both cases, rubisco is not used to fix carbon initially.
 - Both types of plants make most of their sugar in the dark.
- Which of the following statements is a correct distinction between autotrophs and heterotrophs?
 - Autotrophs, but not heterotrophs, can nourish themselves beginning with CO₂ and other nutrients that are inorganic.
 - Only heterotrophs require chemical compounds from the environment.
 - Cellular respiration is unique to heterotrophs.
 - Only heterotrophs have mitochondria.
- Which of the following occurs during the Calvin cycle?
 - carbon fixation
 - reduction of NADP⁺
 - release of oxygen
 - generation of CO₂

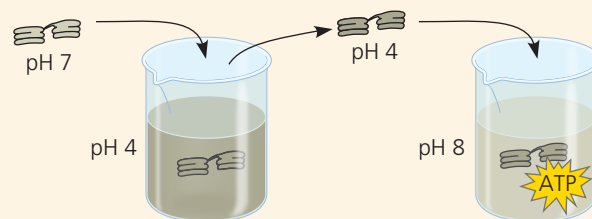
Levels 3-4: Applying/Analyzing

- In mechanism, photophosphorylation is most similar to
 - substrate-level phosphorylation in glycolysis.
 - oxidative phosphorylation in cellular respiration.
 - carbon fixation.
 - reduction of NADP⁺.

- Which process is most directly driven by light energy?
 - creation of a pH gradient by pumping protons across the thylakoid membrane
 - reduction of NADP⁺ molecules
 - transfer of energy from pigment molecule to pigment molecule
 - ATP synthesis

Levels 5-6: Evaluating/Creating

- SCIENCE, TECHNOLOGY, AND SOCIETY** Scientific evidence indicates that the CO₂ added to the air by the burning of wood and fossil fuels is contributing to global warming, a rise in global temperature. Tropical rain forests are estimated to be responsible for approximately 20% of global photosynthesis, yet the consumption of large amounts of CO₂ by living trees is thought to make little or no *net* contribution to reduction of global warming. Explain why this might be the case. (Hint: What processes in both living and dead trees produce CO₂?)
- EVOLUTION CONNECTION** Photorespiration can decrease soybeans' photosynthetic output by about 50%. Would this figure be higher or lower in wild relatives of soybeans? Why?
- SCIENTIFIC INQUIRY • DRAW IT** The following diagram represents an experiment with isolated thylakoids. The thylakoids were first made acidic by soaking them in a solution at pH 4. After the thylakoid space reached pH 4, the thylakoids were transferred to a basic solution at pH 8. The thylakoids then made ATP in the dark. (See Concept 3.3 to review pH.)



Draw an enlargement of part of the thylakoid membrane in the beaker with the solution at pH 8. Draw ATP synthase. Label the areas of high H⁺ concentration and low H⁺ concentration. Show the direction protons flow through the enzyme, and show the reaction where ATP is synthesized. Would ATP end up in the thylakoid or outside of it? Explain why the thylakoids in the experiment were able to make ATP in the dark.

- WRITE ABOUT A THEME: ENERGY AND MATTER** In a short essay (100–150 words), describe how photosynthesis transforms solar energy into the chemical energy of sugar molecules.
- SYNTHESIZE YOUR KNOWLEDGE**



“Watermelon snow” in Antarctica is caused by a certain species of photosynthetic green algae that thrives in subzero temperatures (*Chlamydomonas nivalis*). These algae are also found in high-altitude, year-round snowfields. In both locations, UV light levels tend to be high. Propose an explanation for why this alga appears reddish-pink.

For selected answers, see Appendix A.

Explore Scientific Papers with Science in the Classroom

Can we capture the sun's energy by mimicking plants?
Go to “Replicating Photosynthesis”
at www.scienceintheclassroom.org.

➔ **Instructors:** Questions can be assigned in Mastering Biology.